



MINI PROJECT



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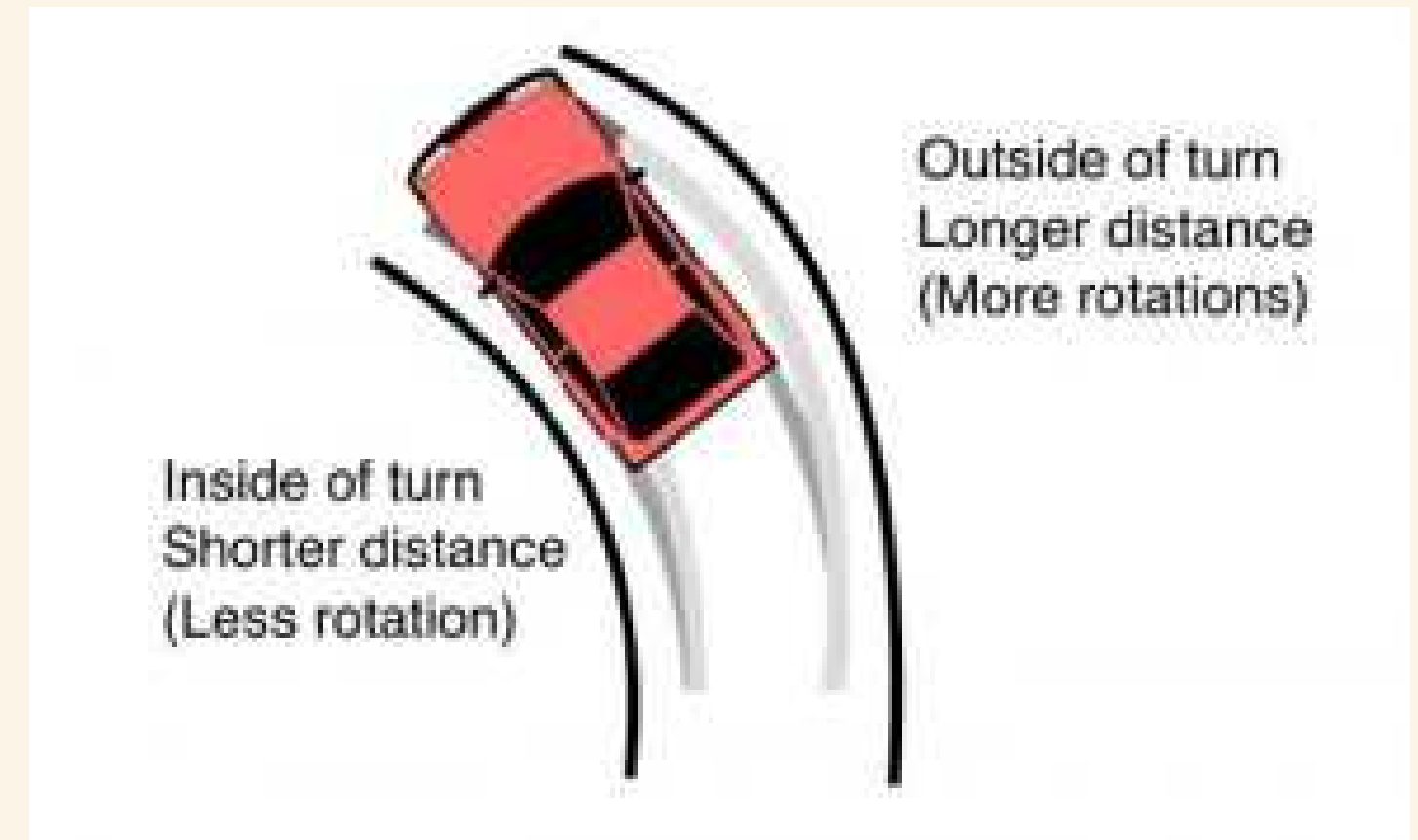
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Engineering Understanding

The Need And the Function

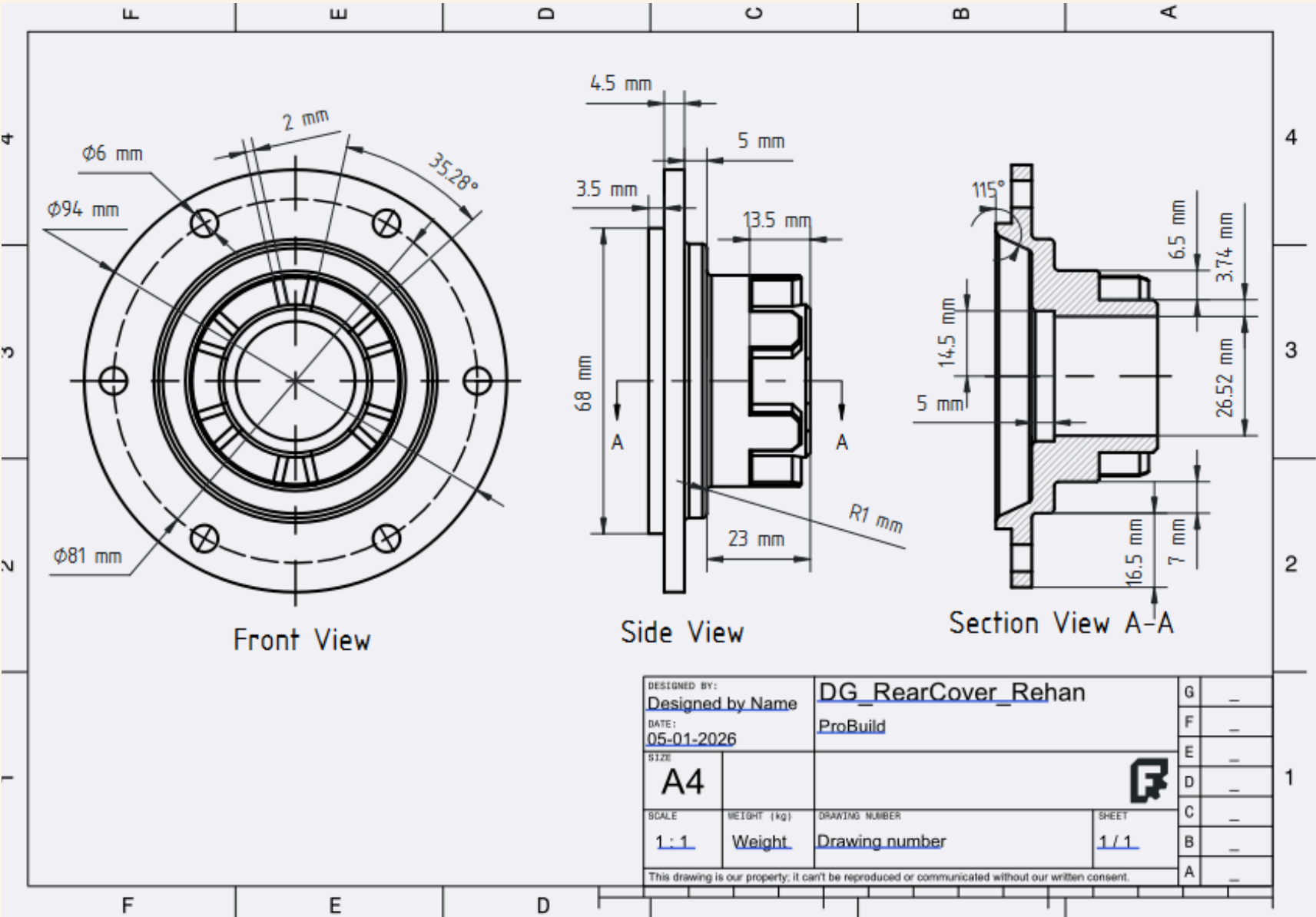
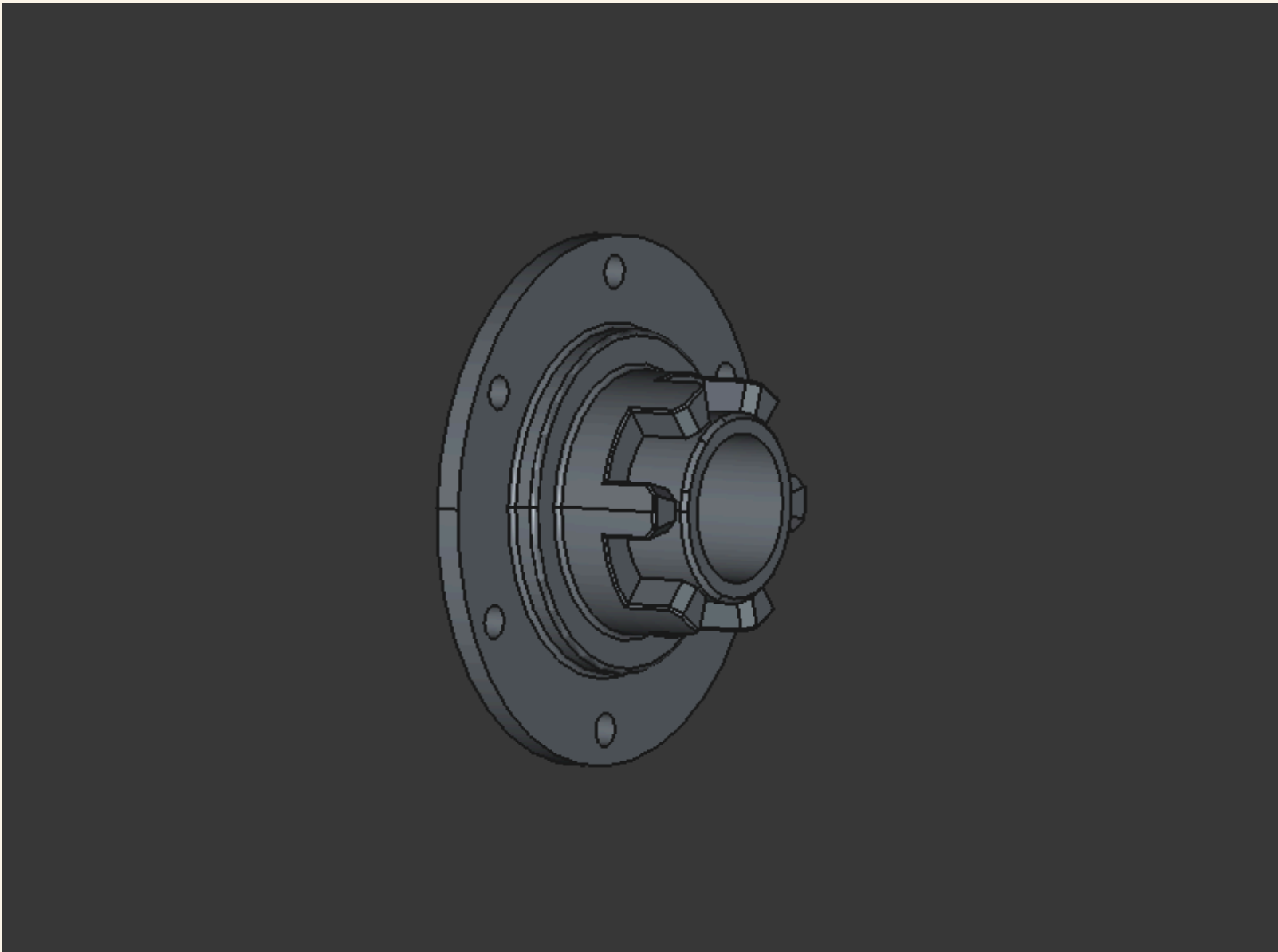
Function of a differential is to allow the drive wheels to rotate at different speeds while still receiving power from the engine.

- The Problem (Turning Geometry): When a vehicle takes a turn, the wheels travel along different paths.
 - The Outer Wheel travels a longer arc (longer distance).
 - The Inner Wheel travels a shorter arc (shorter distance).
- The Consequence without a Differential: If the two wheels were connected by a solid axle (like a go-kart), they would be forced to spin at the same speed. This would cause the inner tire to slip (scrub) against the road, leading to rapid tire wear and difficult steering ("understeer").

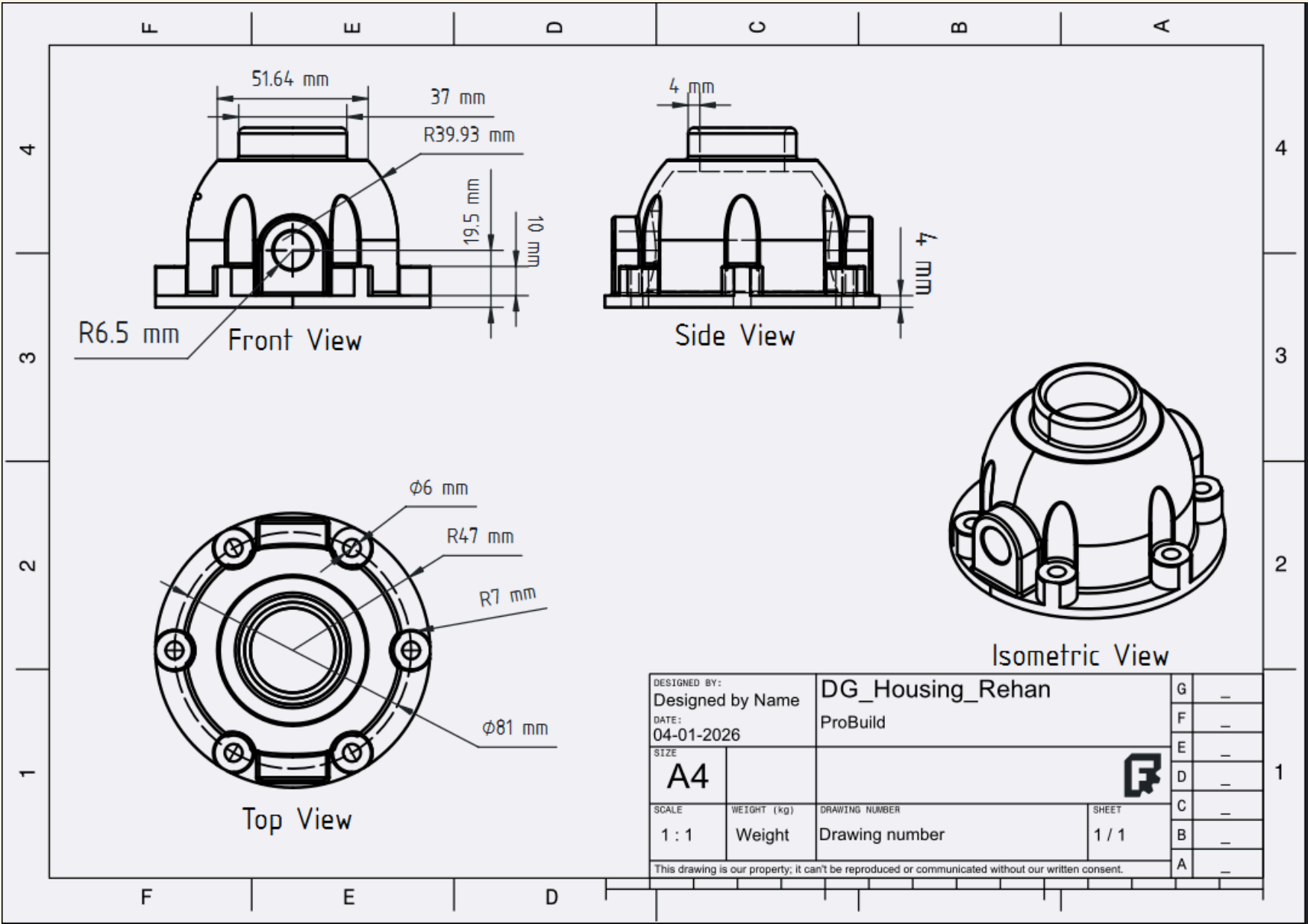
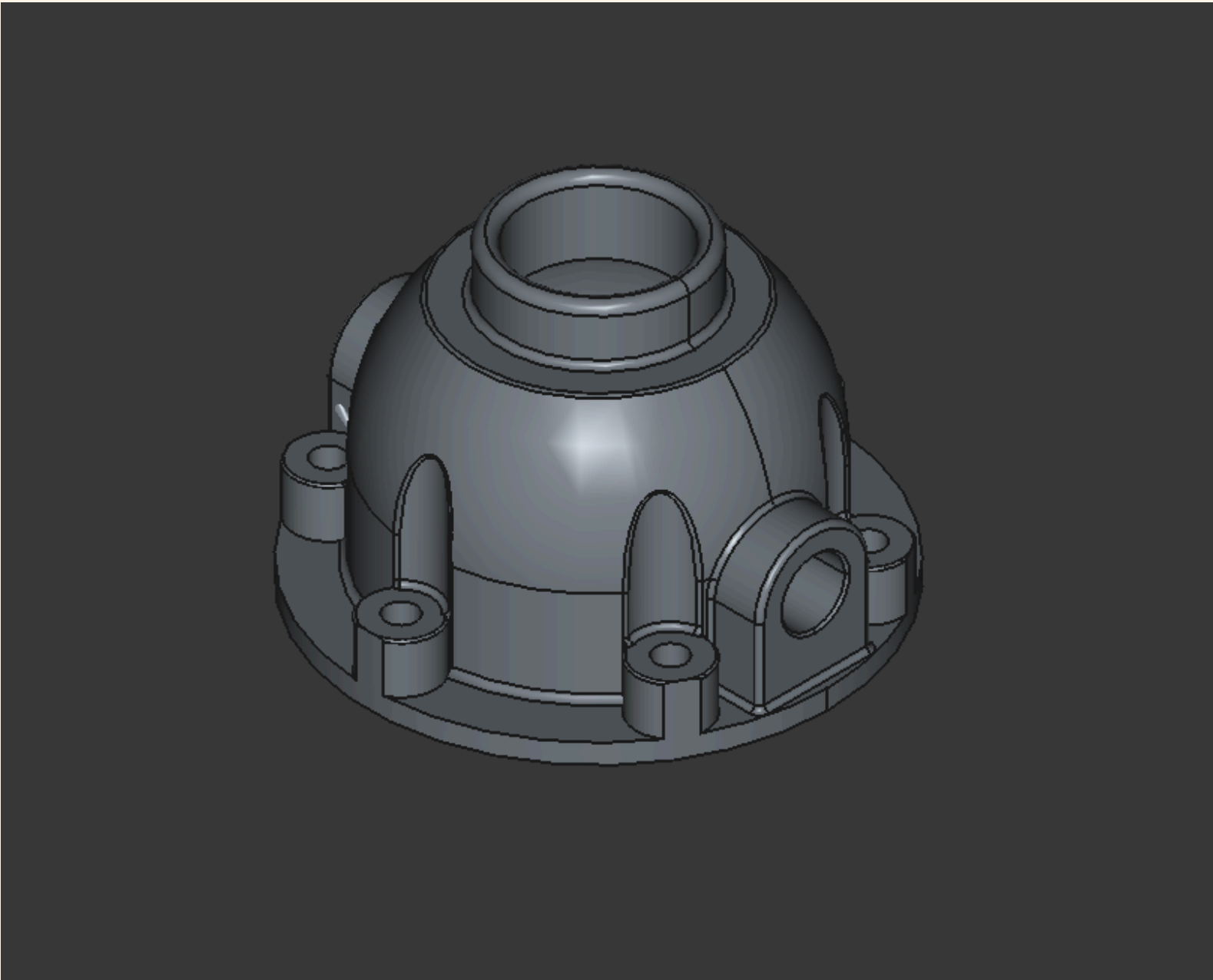


Component's

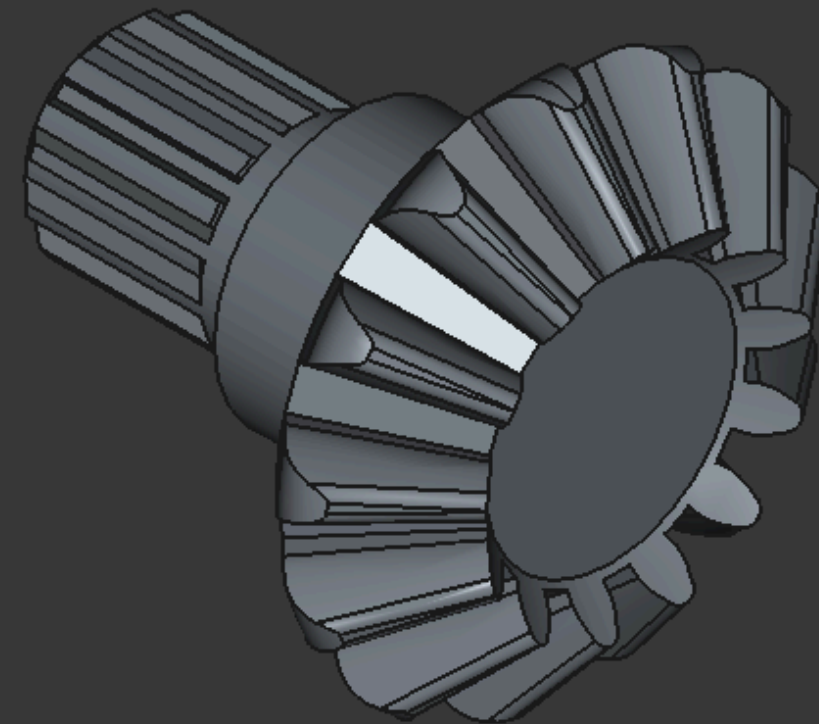
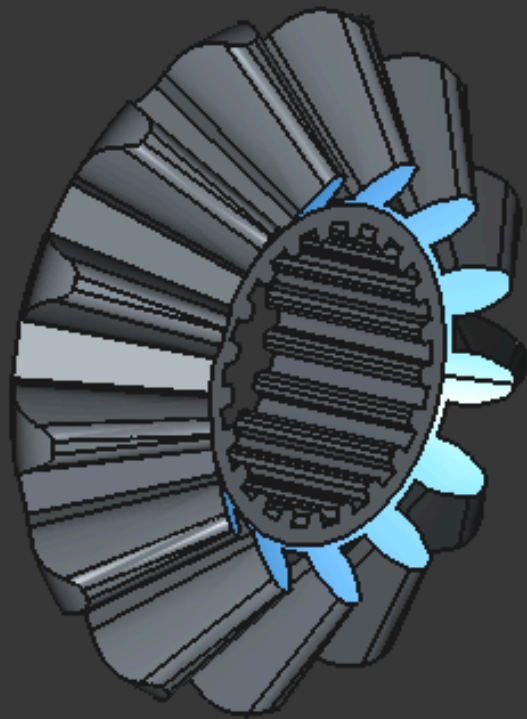
RearCover



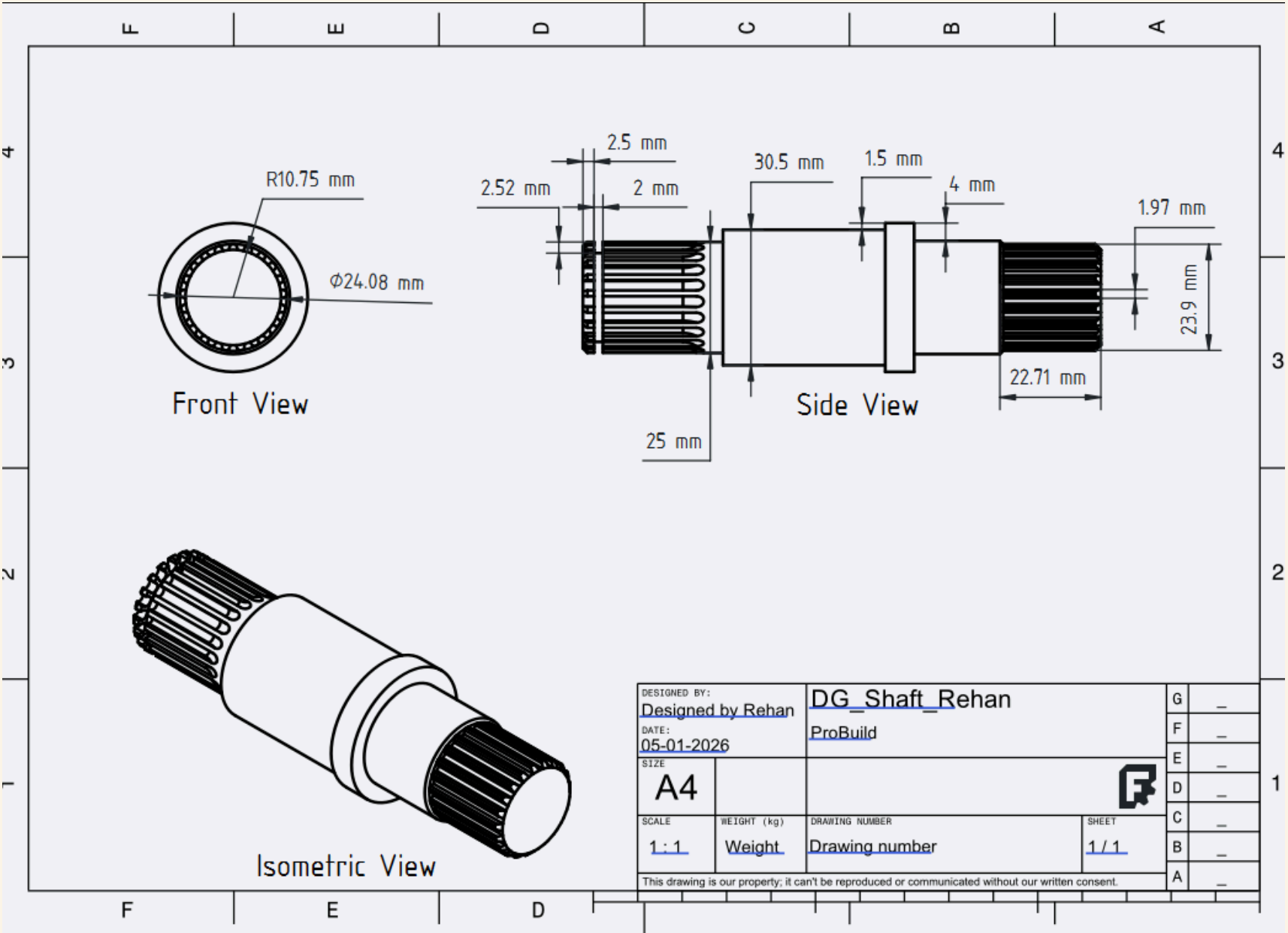
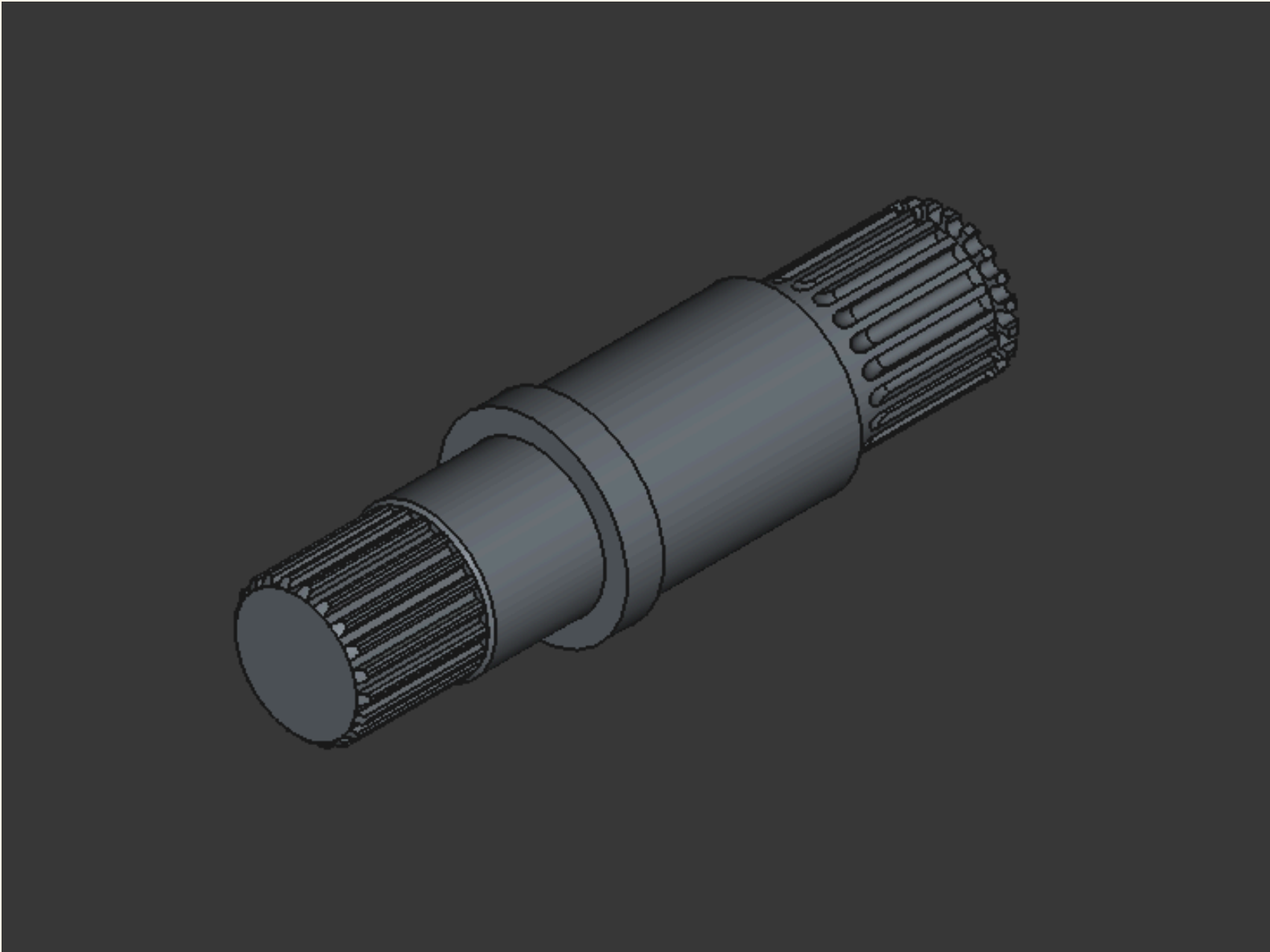
Housing



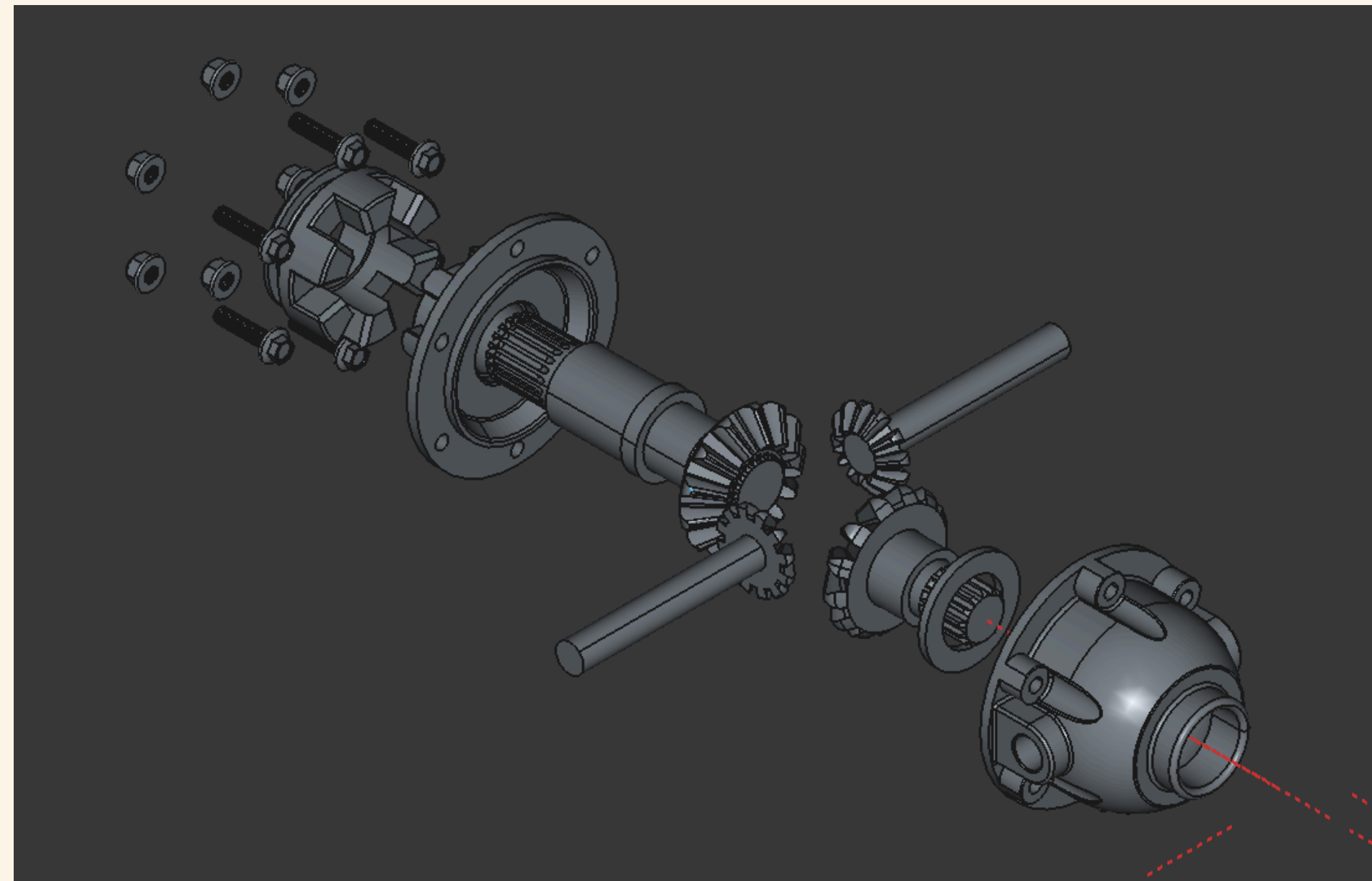
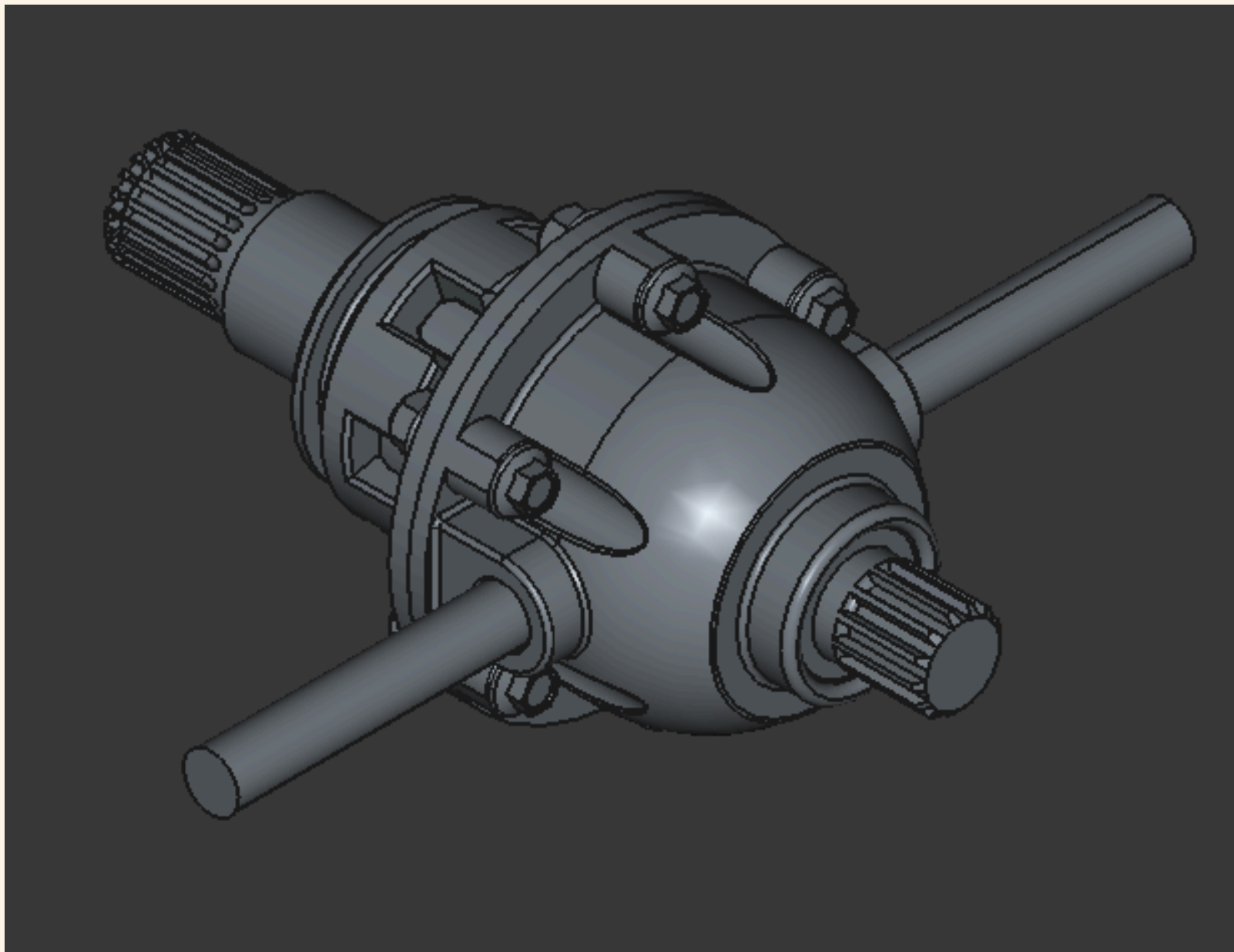
Main Gears



Shaft



Main Assembly



Main Assembly

